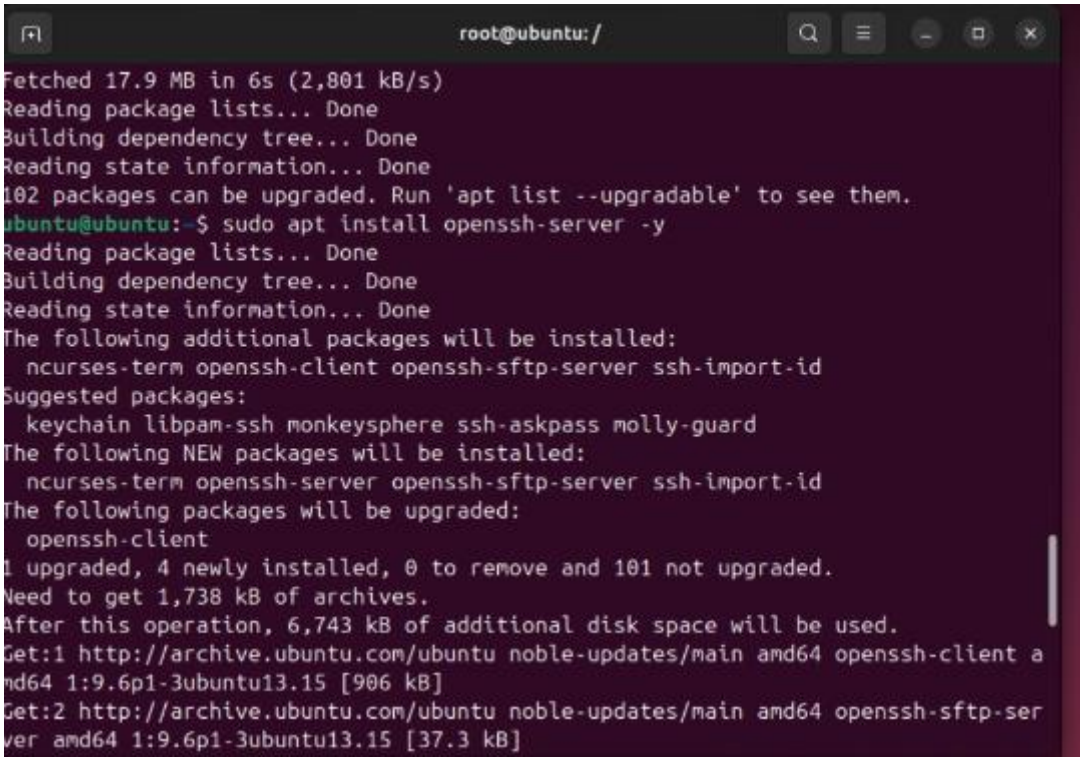
Aluna: Pâmela Layla Freitas Barbosa

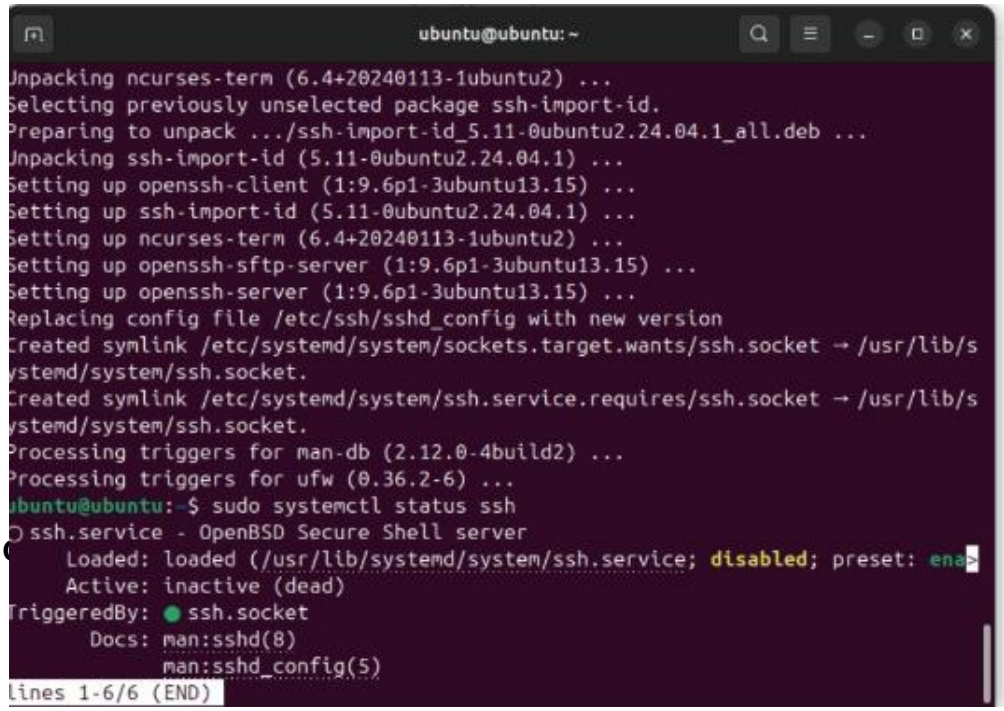
Professor: Ricardo Taveira

Atividade: Avaliação 1

1º: Instalando Openssh-Server:



2º: O SSH funcionando:



3º:

```
Windows PowerShell
O Windows PowerShell
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Instale o PowerShell mais recente para obter novos recursos e aprimoramentos! https://aka.ms/PSWindows

PS C:\Users\Alunos> ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (C:\Users\Alunos/.ssh/id_rsa):
Created directory 'C:\\Users\\Alunos/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Alunos/.ssh/id_rsa
Your public key has been saved in C:\Users\Alunos/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:lA/l4c0f2H1ErvGwJZ5Xc3OEGaSyKpkGk9N7fuCTBCQ alunos@DESKTOP-XXTQZ1G
The key's randomart image is:
+---[RSA 4096]----+
|          o  .o=+|
|    E .  = + +o= |
|     o  + + = *oO|
|     o.. o o o @B|
|    = ..S o   * o|
|     + +o.     . |
|      *ooo       |
|     . ++ .      |
|        .o       |
+----[SHA256]-----+
PS C:\Users\Alunos>
```

4º: Gerando chave pública:

```
PS C:\Users\Alunos\.ssh> type id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDJs+jLimHCc9wRnX4wxUNTK0qkgxcS1EVjkYCsRpKj+FGM78PSHuHPwBsXjMMdZ3aKhOpbSzO6Fz5BQp5P
SF4YDIdKbsiVpCYlHCv2NWGDFx83ECHmHenTaUyBqh0JUTdAqQzndhxnnk8JtCVBpyYCk1aD2jbT8HhFpxULemRQuB5g+foTlMLlkjFFMVbSCabIhwN/e7oE
OvfUYZUmPxrhlJuzU3D0Jmws6xLGhsFocFnXrYiprV281/gEtWtZUWwqUC9EbsrNGBmM6KcdEGpAxIl/pH7wncz1RGS1F/MHg/rNuwzGsvTnxvLNMQMDLwDF
RTUk[illegible]ooXPNbLMrrxAdKAuOYBlACae0juLBohafNNqKwTasniYxpaIlz7YWCnnPIJ8DgbO83OV5yvPb8BqneHabEhLV59Co/Qz0VxswVuIt4X2uhWUre+Opl
CT4H[illegible]krvkD2hn4NyRiqggzJDvzIQySaOADc30zUOm2IsifkRSfz2AQpCRB8oM09TJeBN9q7cRfd9dsTZNq6F7N3WzPyVUf7YgUwTlMV24xXWDC43zcUY294L
gMpAeXLgnLJBO48TXRog6SvRCTxcCqD5sK7zk1IqfkLByqAksnRM9474pgxkeblkywn0Cuf4vyPkDhOEksscpmT0Z9fpO4EaxBlRBuu3wqO10VjQw2X+OM+b
Kw== alunos@DESKTOP-XXTQZ1G
PS C:\Users\Alunos\.ssh>
```

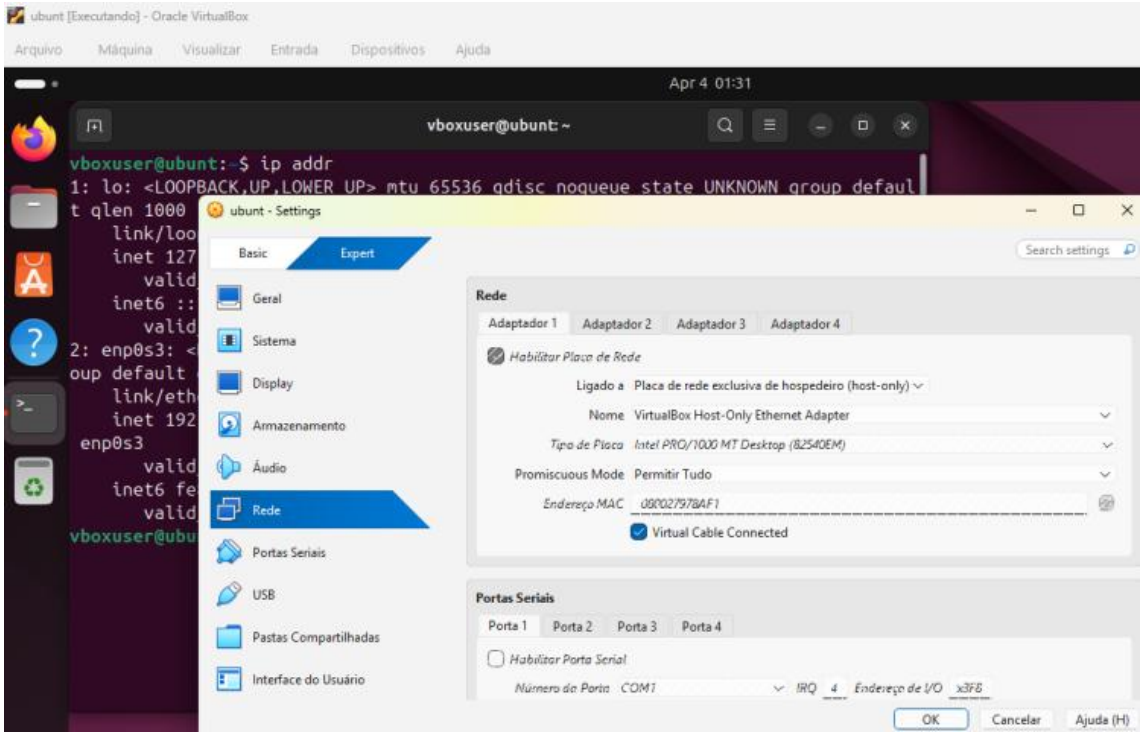
5º: Gerando chave privada:

6º:

```
PS C:\Users\Alunos\.ssh> type id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAdzc2gtcn
NhAAAAAwEAAQAAAgEAybPoy4phwnPcEZ1+MMVDUytKpIMXEtRFY5GArEaSo/hRjO/D0h7h
z8AbF4zDHWd2ioTqW0szuhc+QUKeT0heGAyHSm7IlaQmJRwr9jVhgxcfNxAh5h3p02lMga
odCVE3QKkM53YcZ55PCbQlQacmApNWg9o20/B4RacVC3pkULgeYPn6E5TC5ZIxRTFW0gmm
yIcDf3u6BDr31GGVJj8a4Z5bs1Nw9CZsLOsSxobBaHBZ162Iqa1dvNf4BLVrWVFsKlAvRG
7KzRgZjOinHRBqQMSJf6R+8J3M9URktRfzB4P6zbsMxrL058byzTEDAy8AxUU1JFGqKFzz
WyzK68QHSgLjmAZQAmntI7iwaIWnzTaisE2rJ4mMaWiJc+2Fgp5zyCfA4GzvNzlecrz2/A
ap3h2mxIS1efQqP0M9FcbMFbiLeF9roVlK3vjqZQk+BzpK75A9oZ+DckYqoIMyQ78yEMkm
jgA3N9M1DptiLIn5EUn89gEKQkQfKDNPUyXgTfau3EX3fXbE2Tauhezdlsz8lVH+2IFME5
TFduMV1gwuN83FGNveC4DKQHly4JyyQTuPE10aIOkr0Qk8XAqg+bCu85NSKn5CwcqgJLJ0
TPeO+KYMZHm5ZMsJ9Arn+L8j5A4ThJLLHKZk9GfX6TuBGsQZUQbrt8KjtdFY0MNl/jjPmy
sAAAdQ3O0qV9ztKlcAAAAHc3NoLXJzYQAAAgEAybPoy4phwnPcEZ1+MMVDUytKpIMXEtRF
Y5GArEaSo/hRjO/D0h7hz8AbF4zDHWd2ioTqW0szuhc+QUKeT0heGAyHSm7IlaQmJRwr9j
VhgxcfNxAh5h3p02lMgaodCVE3QKkM53YcZ55PCbQlQacmApNWg9o20/B4RacVC3pkULge
YPn6E5TC5ZIxRTFW0gmmyIcDf3u6BDr31GGVJj8a4Z5bs1Nw9CZsLOsSxobBaHBZ162Iqa
1dvNf4BLVrWVFsKlAvRG7KzRgZjOinHRBqQMSJf6R+8J3M9URktRfzB4P6zbsMxrL058by
zTEDAy8AxUU1JFGqKFzzWyzK68QHSgLjmAZQAmntI7iwaIWnzTaisE2rJ4mMaWiJc+2Fgp
5zyCfA4GzvNzlecrz2/Aap3h2mxIS1efQqP0M9FcbMFbiLeF9roVlK3vjqZQk+BzpK75A9
oZ+DckYqoIMyQ78yEMkmjgA3N9M1DptiLIn5EUn89gEKQkQfKDNPUyXgTfau3EX3fXbE2T
auhezdlsz8lVH+2IFME5TFduMV1gwuN83FGNveC4DKQHly4JyyQTuPE10aIOkr0Qk8XAqg
+bCu85NSKn5CwcqgJLJ0TPeO+KYMZHm5ZMsJ9Arn+L8j5A4ThJLLHKZk9GfX6TuBGsQZUQ
brt8KjtdFY0MNl/jjPmysAAAADAQABAAACAQDIhiXQBNk/unN0x8BmuBZAnu9sMi9ENuht
HoBvxRGQFhTtKojhYnZYVcIZXYW0u+FrwMxEyrS4Mtdfhd1hMK7q9bmi1BRyQGjzx9gkw9
kXvGr6EP3CisaluS+oUPjhBG8Lx4Ter9t5YDdvjDrl1EjDkrsWBmEvvwezJ3KSoLEOVp1N
B/gIc4J3PSnigjSwR4waG6/sk5LRFdbAu+rNxTryPQ218CSBdZlh5qoZ2Gp7bAbFlu4g0g
w46/TUqY/5kisYvfLf882GrIjqP3m0OVeI4FtaHK1ehWL4C3CbYR3pwNocofMHr24u5ukH
+cvm5Fm9xRoZ615hFleRsyxhy9b5hoTk9tvTp+HZezcOW9W7N6RBzzM/CpzzxtrH5/jU8k
U62aitRL3A/5GUP3BTtFtoEsGN1CBXSlddZTuk9mKcg6Y5Wcv9vyBZP5EyAblMrCtCjiQz
iWlIFima3ReiXm2TcqMbLK7w746reUSJXOIDaFB7i699cEhHeeQrYXV/VFw7nUzYWkKe5K
ZKioMoATGILNXTOicbFJtVH4ApHWMnglwj2ojzGtOYT6DDuez8seYSlV68O6LVKlcbsJpd
vGytXuS65BktMm5VHIrZ83TIE1bTxyxT5N0XyCAsWpoVPkipXpmRSxi/JOKMSbJWYBU250
2fyzU+ruY92F3CDXzF0QAAAQEAg2ZNQLvSs87hZHAP6Xn81UfelV1OKhWLTJCwotDveMLv
LCN2NbyFn89fodP0Z0F+/9d3W9q1DjiEyxvFIs/8JdzcNr7iCGBScerJdeQyk1QXPsWmic
/bKzyH0bP06aG96eOIq8w0psF5YneCnU2NRVq6lUhy/+0oTA9IulWeh4B8Vr9dhA1RLxZP
BOe2BsNVcYnD8tnm2OxUa4vP9vgOcy6pItOkGwlPELsVj3dFv8nxH12nq/fMaV+XYm0Ct0
s9zz8fU83xNYX8gZzh68vldp/oy5QgM0Maf+MjHFFvq5errrJ56BTkS9MsgTbmZYNrA7lI4
DmSH4GfCSrMi5ewuUQAAAQEA8+vsVJr/EkKBPKC11fzgVR/8pBPsvpWxP7InJmlAGIS3e+
8THpxa66TNtm1it5zXg0Vkzoto2eqj/lZ4aLkLjQXsHo1FQm87nGYYPB+g47w/AbBJPpR5
BmTFJgdbid8awKS5na5syWJzu220fjkl+bMwynNFs9NSDDrItT1KVD7g6hRNCdUaE4reiX
6SEBfeIhAfZvO8qA2VFEqUbv2omXj0TWrYOqoffq0kOpULsJhFxpmVyn15x6dlzECP8Wyc
6TdbkK0h2D8dj2JCfiGM5tz+szsHtjNm+igmXiWIOsByHECyLegh4SjzSjxGEjVz+5zX9q
HVEBCd8gUhGnzY/QAAAQEA07DMdNdmI2EbZ25gY/njqF/4Yq5dEkEaxCyHMYZP2rlpfXcb
Cuc79yXgtobiDisZC7zlsmpNib9UHiG5X9YYVGPK4k3pAAFxZgwT1TEj5/W5lJ2jAqth4F
bCj3Vz+9U6ioVHuNl50B7lr6Zg4BbVby/84Xz2m6MKhQ0eF1TgP8cziCzQouVKYaW+UMfU
Pq7ZcK1Vd7FMb1pmlhdcj9pjuefctI9DgpOiMl2dqZUj4bl/oLp7+bYcDA7SDKgqeg6pwo
cmz+VBec2Tkb86aKWmHio475WSpZOKbZhpFxuruCyne6UYY1c6L7uFg8IMeLLvC6bb/X6v
GQo5mOL21XMxRwAAABZhbHVub3NAREVTS1RPUC1YWFRRWjFHAQID
-----END OPENSSH PRIVATE KEY-----
PS C:\Users\Alunos\.ssh>
```

```
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~        ubuntu@ubuntu: ~
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
101 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/system
d/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.s
ervice.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/l
ib/systemd/system/ssh.service.
ubuntu@ubuntu:~$ sudo systemctl start ssh
ubuntu@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
     Active: active (running) since Wed 2026-03-18 00:31:29 UTC; 6s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 5690 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5692 (sshd)
      Tasks: 1 (limit: 2252)
     Memory: 1.3M (peak: 1.5M)
```

7º: Configuração de Rede no VirtualBox:



```
vboxuser@ubunt: ~
boxuser@ubunt:~$ ip addr
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
 qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
up default qlen 1000
    link/ether 08:00:27:97:8a:f1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute
 enp0s3
       valid_lft 554sec preferred_lft 554sec
    inet6 fe80::a00:27ff:fe97:8af1/64 scope link
       valid_lft forever preferred_lft forever
boxuser@ubunt:~$
```

8. Teste de Conexão Final

```
vboxuser@ubunt: ~
PS C:\Users\RED>
PS C:\Users\RED> ssh vboxuser@127.0.0.1 -p 2223
kex_exchange_identification: read: Connection reset
Connection reset by 127.0.0.1 port 2223
PS C:\Users\RED>
PS C:\Users\RED> ssh vboxuser@192.168.56.101
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:N7mB9Xoa0OhZ7FvEGqQNGXvxlvXtjhZpHGq3Ym0iImQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.
vboxuser@192.168.56.101's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-20-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Apr  4 00:56:20 2026 from ::1
```

```
vboxuser@ubunt:~$ Get-Content "$env:USERPROFILE\.ssh\id_rsa.pub" | ssh vboxuser@192.168.56.101 "mkdir -p ~/.ssh; cat >>
~/.ssh/authorized_keys"
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:N7mB9Xoa0OhZ7FvEGqQNGXvxlvXtjhZpHGq3Ym0iImQ.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? Get-Content: command not found
yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.
vboxuser@192.168.56.101's password:
vboxuser@ubunt:~$
```